

DATA AT THE USDA

US Department of Agriculture
Local and Regional Foods Division: Research Branch
Annelise Straw— Agricultural Marketing Specialist

QUALITATIVE ANALYSIS WITH COPILOT

US Department of Agriculture | Local & Regional Foods Division (LRFD)

Nayab Safdar, Data Fellow | *University of Maryland, M.S. Business Analytics & AI*

Keywords:

Qualitative analysis, tracking common themes, prompt engineering

Summary:

To assess materials from **17 different interviews** involving two focus groups, Nayab utilized prompt engineering to collaborate with **Microsoft Copilot** to identify common recurring themes, cross division challenges and opportunities, as well as recommendations and trends specific to each program. Compiling the structured analysis into a shared doc with the prompt used would allow leadership to make any changes or replicate the process with other interviews.

DATA CLEANING AND RESHAPING

US Department of Agriculture | Local & Regional Foods Division (LRFD)

Nayab Safdar, Data Fellow | *University of Maryland, M.S. Business Analytics & AI*

Keywords:

Data cleaning, reusable pipelines, interactive visualizations, text parsing

Summary:

To evaluate and clean the **20,000+ rows** of Local Agriculture Market Program (LAMP) indicator data from 2024, Nayab split the data into structured columns using **Regex**, calculated descriptive stats, and removed non-standard rows through EDA. Using **Python and Power BI**, Nayab built an interactive dashboard and a reusable a data cleaning pipeline that could be applied to other program data, ultimately turning messy raw data into a clean, presentable table. This structured process would ultimately cut down data processing time by **1+ years** and produce validated datasets that could then be fed into internal/external dashboards.

INDICATOR FREQUENCY ANALYSIS

US Department of Agriculture | Local & Regional Foods Division (LFRD)

Nayab Safdar, Data Fellow | *University of Maryland, M.S. Business Analytics & AI*

Jackie Liang, Data Fellow | *University of Pennsylvania, MPA*

Keywords:

Data cleaning, frequency analysis, multi-program comparison

Summary:

To map out the frequency at which each program-specific evaluation survey question (outcomes/indicators) is being reported on across **4 different programs**, Nayab normalized and reshaped the data before adding in an already mapped schema breakdown separating old indicators from new ones, identified through text parsing. Using **Python**, Nayab created frequency tables tracking the usage of only new indicators for 2022 and onwards. The product is a clean table providing counts of how many times an indicator was reported on across the different programs to help leadership with **pruning unpopular indicators** off the total 1,800+ indicators.

The Local and Regional Foods Division (LRFD)
leads purpose-driven research and delivers
practical tools that help farmers and food
businesses access high-value markets,
improve profitability, and overcome barriers
to growth.

DATA CLEANING AND RESHAPING

Local Agriculture Market Program (LAMP) indicator data from 2024

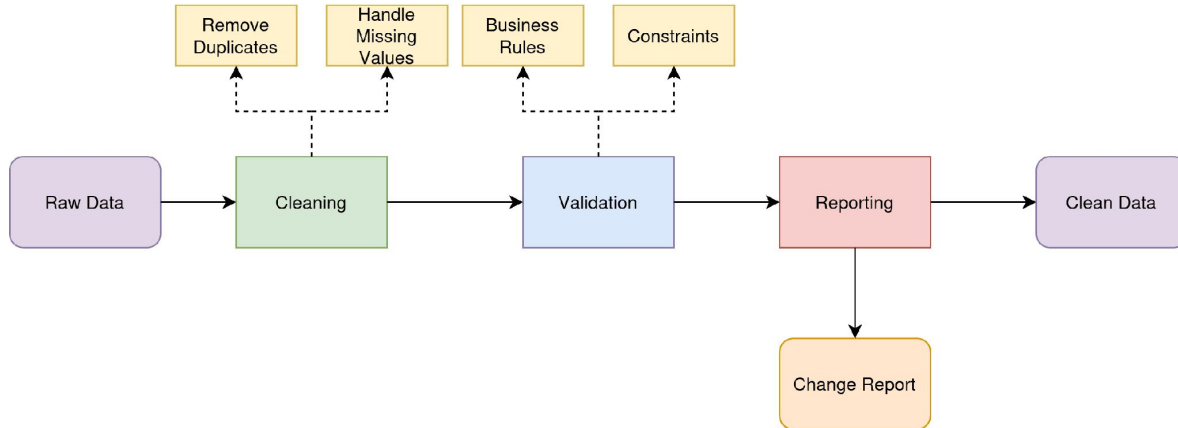
CONTEXT

- The Agricultural Marketing Services (AMS) Grants Division funds projects that enhance marketing opportunities for U.S. farmers, producers, and local food systems.
- **Key Grant Programs:** Local Agriculture Market Program (LAMP), Specialty Crop Block Grant Program (SCBGP), and Resilient Food Systems Infrastructure Program (RFSP).
- Collect progress performance survey data (outcomes/indicators) from these programs and store it in a database



PROCESS

- Created consistent formatting for the dataset to clearly analyze standard vs. non-standard indicator values
- Split the data into structured categories using keywords for easier data navigation



DELIVERABLES

- Built a documented Jupyter Notebook to handle data cleaning, reshaping, formatting, and exporting which can also be used for other programs.
- Created a Power BI dashboard to visualize LAMP indicators and outcomes, making it easier to explore trends, distributions, and which indicators are popular.



INDICATOR FREQUENCY ANALYSIS

4 AMS grant programs; analysis only on indicators currently in use

PROGRAM DATA

- The grantees awarded by the Agricultural Marketing Services (AMS) are required to provide Progress Performance Reporting (PPR) data -- selecting different indicators that measure different outcomes, unique for each program.
- Since each program has distinct indicators and outcomes, this brings the division to thousands of unique indicators, a lot of which are unused and need to be removed for a smoother data collection process.



11+ programs

Are being currently managed by the AMS Grants Division

PROJECT DETAILS

Step	What was done	Why
1	Loaded the provided PPR datasets and derived the award year from the award number	Establishes the time window the analysis needs to cover and creates a new traceable metric.
3	Built an exploratory check comparing each program's set of outcome names year over year, to see where indicators had changed.	The program indicators are known to have shifted around 2022; this made the scale of that drift visible before deciding how to handle it.
4	Merged in the “old”/”new” schema indicator classification data from an external reference file, validated the merge, and filtered to currently used data only.	Closes the gap the exploratory check identified. The programs now have a standard “old”/”new” indicator categorization.
5	Restricted the four-program analysis to include those with final reports only, then ran the unique-grantee-per-indicator count, split by program and grouped by outcomes.	Produces a structured deliverable across all four programs with raw counts and percentages tracking each indicator response.
6	Exported the results to a second formatted Excel workbook.	Helps populate the team’s mapping Excel document to track indicator performance for each program.

DELIVERABLES

- Tracked and calculated the frequencies of each valid indicator across 4 programs to help the team populate internal records and assess outcome/indicator priority to inform future pruning decisions to ensure efficient data collection practices.
- Developed a Power BI dashboard to visualize descriptive statistics, indicator distribution across programs, locations, and organizations, response rates per indicator, and program-level outliers for internal use as a single source of information about all the related awards.



Questions?

Thank you to **Annelise Straw, Elliot Hohn, Lacy Stephens, and Cassie Rubio** for supporting me throughout this experience and making it memorable!